# NVIDIA Cosmos, Explained in Depth

A beginner-to-intermediate guide to all four generations — Cosmos 1, 2, 2.5, and 3 — with diagrams, model sizes, data, inference speed, and the pretraining + post-training (SFT / GRPO) recipes.

## What is a "World Foundation Model" (WFM)?

A **language** model learns from text and predicts the next *word*. A **World Foundation Model** learns from video and predicts what happens next in the *physical world* — how things move, collide, and behave. NVIDIA **Cosmos** is a family of WFMs for **Physical AI** (robots, self-driving cars): instead of testing a robot millions of times in reality, engineers let Cosmos **imagine** realistic video of what would happen and train on that.

> *Mental model: Cosmos is a "dream engine" for machines. Each generation dreams with higher quality, runs faster, understands your request better, and — by Cosmos 3 — even hears sound and decides physical actions.*

### Mini-glossary (so the rest makes sense)

| | |
|---|---|
| **Tokenizer** | Compresses video into a small set of numbers ("tokens") the model can handle, and decompresses them back to pixels. |
| **Diffusion** | Starts from random noise and repeatedly "cleans" it into a video. High quality, but many steps = slower. |
| **Autoregressive (GPT-style)** | Generates the video one chunk at a time, like typing word by word. Fast, good for real-time. |
| **Flow-based** | A cleaner, more direct cousin of diffusion: learns a straight "flow" from noise to data, often with fewer steps. |
| **DiT** | "Diffusion Transformer" — the Transformer architecture that powers the diffusion engine. |
| **Mixture-of-Transformers** | Several specialized Transformers in one model (e.g. one that *reasons* + one that *generates*). |
| **Pretraining** | Learning general knowledge from a huge, broad dataset. |
| **SFT** | Supervised Fine-Tuning: teach the model on curated question→answer examples after pretraining. |
| **GRPO** | An RL method: the model tries an answer several times; better-than-average tries get reinforced. Sharpens reasoning. |

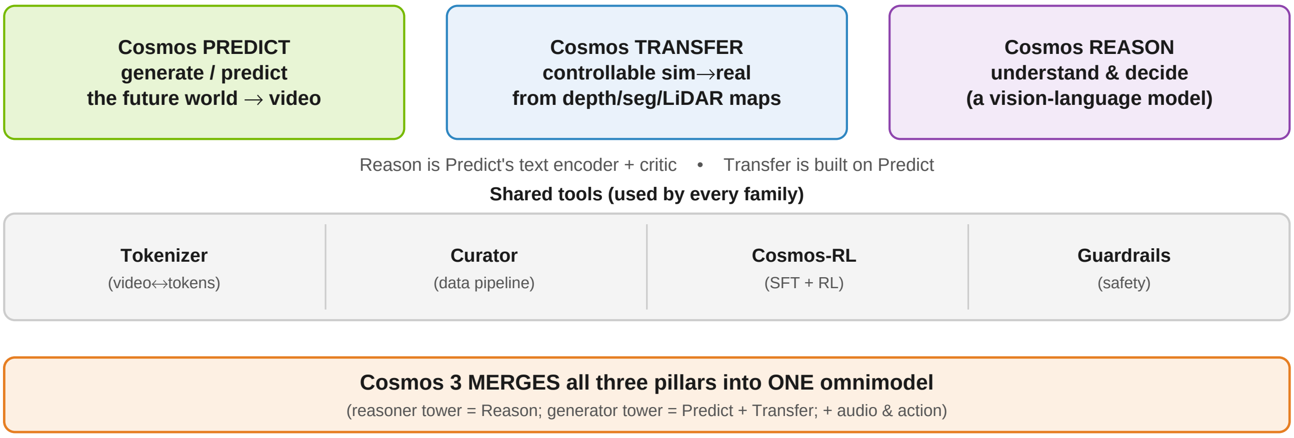
## The big picture: all four generations side by side

| | Cosmos 1 | Cosmos 2 | Cosmos 2.5 | Cosmos 3 |
|---|---|---|---|---|
| **Released** | Jan 2025 | Jun 2025 | Sep 2025 | Jun 2026 |
| **Core engine** | TWO engines:<br>diffusion + GPT-style | Refined diffusion | ONE unified<br>flow-based model | Mixture-of-Transformers<br>(reason + generate) |
| **Inputs** | Text, or image/video | Text, or image/video | Text/image/video<br>(one model) | Text, image, video,<br>AUDIO + ACTIONS |
| **Outputs** | Future video | Video (480/704p) | Video up to 30s,<br>multi-camera | Video + actions +<br>audio, with reasoning |
| **Reads text via** | T5-XXL encoder | T5-style encoder | Cosmos-Reason1 (VLM) | Built-in reasoning<br>transformer |
| **Model sizes** | Diff 7B/14B,<br>GPT 4B/12B | 0.6B / 2B / 14B | 2B / 14B<br>(+auto/robot variants) | Super / Nano / Edge |
| **Training data** | ~100M clips<br>(from 20M hrs) | + action datasets | 200M curated clips | ~20 trillion tokens |
| **Post-training** | Video2World<br>fine-tune | Action-conditioned<br>fine-tune | Model merge + RL | SFT + RL (GRPO-style) |
| **One-word vibe** | Kitchen sink | Polish | Unify | Leap |

Table 1. Each column is colour-coded and reused throughout this guide: Cosmos 1, Cosmos 2, Cosmos 2.5, Cosmos 3.

## Important: Cosmos is a PLATFORM, not a single model

So far we compared “generations” (1, 2, 2.5, 3). But each generation is really a **bundle of separate model families that do different jobs**. The autoregressive 4B model, for example, is just one variant inside the **Predict** family of generation 1. There are three core “pillars” plus shared tools.




Platform map. Three pillars (Predict / Transfer / Reason) sit on shared tools. They interconnect — and by Cosmos 3 they fuse into a single model.

### The model families at a glance

| Family | Its job | Input → Output | Technique | Versions |
|---|---|---|---|---|
| Predict | Generate / predict the future world | text/image/video → video | diffusion + AR (v1) → diffusion (v2) → flow (v2.5) | 1, 2, 2.5 |
| Transfer | Controllable generation; sim→real | structure maps (depth, seg, edge, LiDAR, HD-map) + text → photorealistic video | (Multi-)ControlNet on top of Predict | 1, 2.5 |
| Reason | Understand & decide | image/video + text → text reasoning, decisions, (v2) 2D/3D detections | vision-language model (VLM) | 1, 2 |
| Tokenizer | Compress video ↔ tokens | video → continuous/discrete tokens | causal autoencoder (FSQ) | 1 |
| Curator | Prepare training data | raw video → clean, captioned dataset | Ray GPU pipeline | — |
| Cosmos-RL | Train / fine-tune any family | checkpoints + data → fine-tuned models | SFT + RL | — |
| Guardrails | Safety | prompts/outputs → filtered content | classifiers | — |

Table 1a. Predict, Transfer and Reason are different MODELS for different jobs — not versions of one another. Generations 1/2/2.5 ship them separately; Cosmos 3 unifies them.

## Diversity inside ONE family: Cosmos Predict 1

To show how much sits inside a single family-and-generation: **Cosmos Predict 1 alone** is about a dozen distinct checkpoints. The “autoregressive 4B” described earlier is just one of them.

| Variant | Type | What it does |
|---|---|---|
| Diffusion Text2World 7B / 14B | diffusion | text → video |
| Diffusion Video2World 7B / 14B | diffusion | image/video → future video |
| Autoregressive 4B / 12B | GPT-style | video → next frames (the one described earlier) |
| Autoregressive 5B / 13B Video2World | GPT-style | text + video → future video |
| Tokenizer CV / DV (×3 rates each) | tokenizer | compress / decompress video |
| Prompt Upsampler 12B | LLM | expand short prompts into rich captions |
| Diffusion Decoder 7B | diffusion | clean up the autoregressive output |

Table 1b. One family, one generation ≈ a dozen checkpoints. Multiply by Transfer + Reason + tools to see why “Cosmos 1” is a platform, not a model.

### How the families work together

- **Reason → Predict:** Reason1 is reused as the **text encoder** inside Predict 2.5, and as a **quality critic** during data curation.
- **Predict → Transfer:** Transfer 2.5 is **built on top of** Predict 2.5 — a ControlNet wrapped around it so you can steer generation with depth/segmentation/LiDAR maps.
- **Transfer's special role:** it doesn't invent freely; it **converts a structured input** (e.g. a simulator's segmentation video) into photorealistic video — the **sim-to-real** and data-augmentation workhorse.
- **Tokenizer / Curator / Cosmos-RL / Guardrails** feed and support every family.

*Takeaway: don't read Cosmos as one model getting bigger. Read it as a platform of specialised families (Predict = imagine, Transfer = restyle/control, Reason = think) that generation 3 finally fuses into a single omnimodel.*

## Each family evolved on its OWN cadence (not in lockstep)

A subtle but important point: Predict, Transfer and Reason did **not** all version together. Predict went 1→2→2.5; Transfer jumped 1→2.5 (no public “2”); Reason has its own 1→2 track. The “generation” labels really follow **Predict**. And each generation is **not** equally “diverse”:

| Era | Predict | Transfer | Reason | How diverse? |
|---|---|---|---|---|
| Cosmos 1 (2025 H1) | Predict 1 | Transfer 1 | Reason 1 | Full platform |
| Cosmos 2 (mid 2025) | Predict 2 | — | — | Mostly a Predict-only refresh |
| Cosmos 2.5 (Sep 2025) | Predict 2.5 | Transfer 2.5 | Reason 1 reused (Reason 2 later) | Full platform again |
| Cosmos 3 (Jun 2026) | → generator tower | → folded in | → reasoner tower | Unified into ONE model |

Table 1c. Cosmos 2 was a Predict upgrade; 2.5 restored the full platform; Cosmos 3's novelty is that it STOPS being a platform of separate models and fuses them.

### The three families converging into Cosmos 3

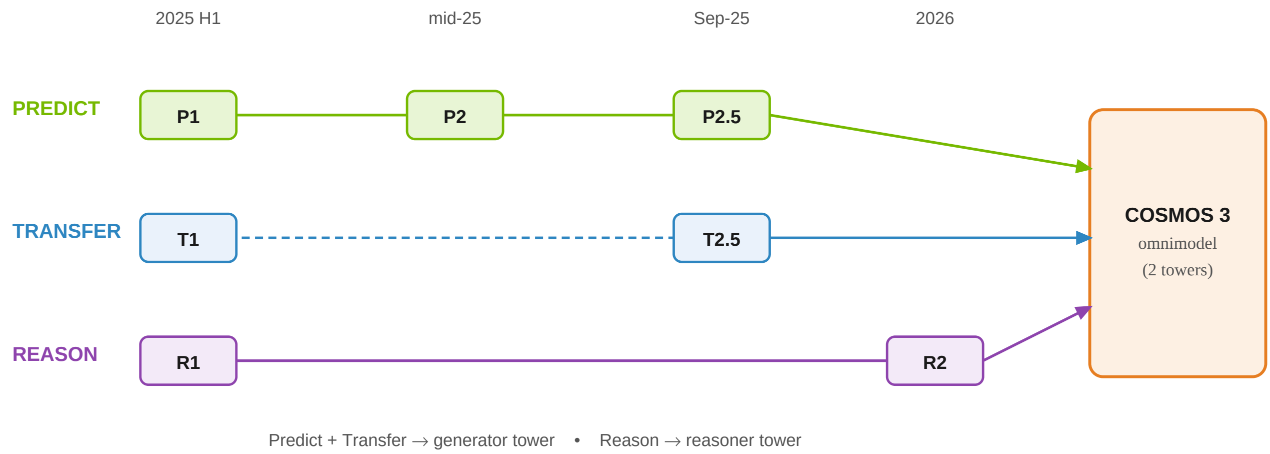



Figure 2a. Dashed Transfer lane = it skipped a public “2”. All three families converge into Cosmos 3's two-tower omnimodel.

## Same family, across versions: the novelty at each step

| Family | Evolution (version → novelty) | Fate in Cosmos 3 |
|---|---|---|
| **Predict** | 1: two engines (diffusion 7/14B + autoregressive 4/12B), T5 encoder, separate tasks. 2: diffusion-only DiT (2/14B) + sparse attention → up to 2.6× faster. 2.5: flow-based + unified Text/Image/Video-to-World + Reason1 encoder + RL + 30s multiview. | Becomes the generator tower |
| **Transfer** | 1: Multi-ControlNet on Predict 1; controls = depth/edge/keypoint/segmentation/LiDAR/HD-map; 4 control blocks at the START of the network; 7B. 2.5: built on Predict 2.5; control blocks DISTRIBUTED (one every 7 blocks); 2B is 3.5× smaller than Transfer1-7B with better alignment & less hallucination. | Folded into the generator tower (conditioning / control) |
| **Reason** | 1: physical-AI reasoning VLM (7B/56B), hybrid Mamba-MLP-Transformer, 16K context, SFT + RL (GRPO), long chain-of-thought. 2: 256K context (16× longer), adds OCR + 2D/3D point localization + bounding boxes + stronger spatial grounding; #1 open model. | Becomes the reasoner tower |
| **Tokenizer** | 1: causal autoencoder; continuous (CV) for diffusion + discrete (DV, FSQ 64k vocab) for autoregressive; up to 12× faster than prior tokenizers. | Reused as the visual/audio encoder family (ViT/VAE) |

Table 1d. Read each row left→right to see one family's novelty trajectory, and the right column for where it ends up inside Cosmos 3.

### The one-paragraph synthesis

Cosmos 1 and 2.5 are **full platforms** (Predict + Transfer + Reason + Tokenizer); Cosmos 2 was mainly a **Predict** efficiency/quality upgrade (sparse attention); Reason ran its own **1→2** track (long context + spatial grounding). The big architectural story of **Cosmos 3** is not a new family — it is the **disappearance** of separate families: the generator tower absorbs Predict + Transfer, the reasoner tower absorbs Reason, and audio + action are added on top.

# How the architecture changed (diagrams)

## Cosmos 1 — two parallel engines

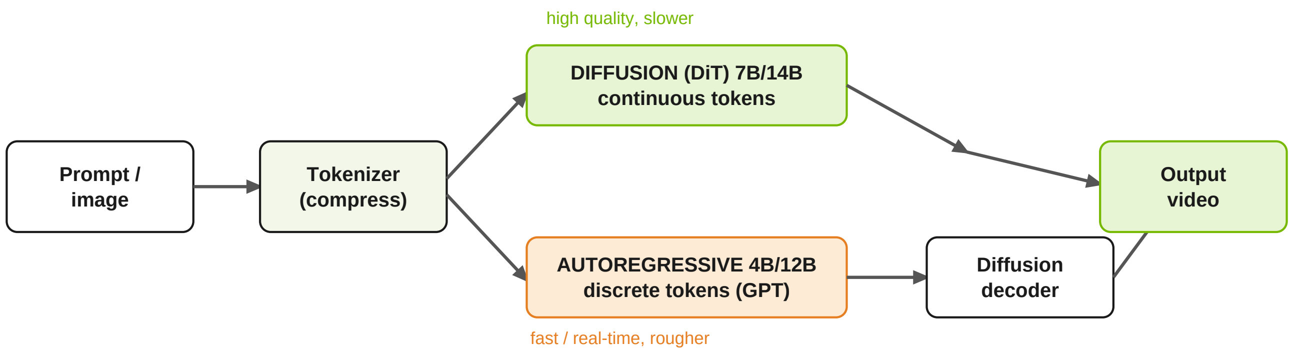



Figure 1. Cosmos 1 shipped two separate engines. Diffusion (top) = best quality; autoregressive (bottom) = fast, but needs a diffusion decoder to clean up its output. Text enters both via a T5-XXL encoder (not shown).

## Cosmos 2.5 — one unified flow model with a “smart” text encoder

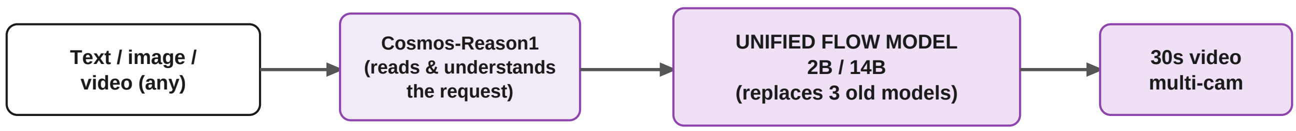



Figure 2. Cosmos 2.5 merged the three separate task-models of earlier versions into a single flow-based model, and upgraded the text encoder to Cosmos-Reason1 — a model that actually understands physics and language.

## Cosmos 3 — mixture-of-transformers (thinks before it dreams)

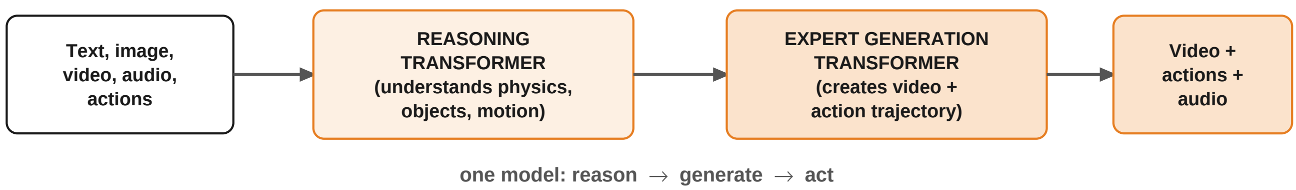



Figure 3. Cosmos 3 is no longer just a video predictor. A reasoning transformer first understands the scene, then an expert generation transformer produces video AND physical actions (and audio) — all in one “omnimodel”.

## Deep-dive: why did Cosmos 1 have TWO engines?

This is the most confusing part of Cosmos 1, so it deserves a clear answer. The two engines were **not** a duplicate or a mistake — the paper states it built them on purpose: *"We explore two scalable approaches for building pre-trained world foundation models — the diffusion model and the autoregressive model."* Both solve the **same** hard problem (predict believable future video) with the same trick — break one hard generation into many easy sub-steps — but in two different "spaces", which forces two different training recipes.

> *Analogy: same destination, two routes. Diffusion is a **sculptor** — start from a rough noisy block and refine the whole video over many passes. Autoregressive is a **writer** — produce the video one "token" at a time, left to right, like an LLM.*

| Aspect | Diffusion engine (sculptor) | Autoregressive engine (writer) |
|---|---|---|
| Token space | Continuous latents (CV 8×8×8 tokenizer) | Discrete codes (DV 8×16×16; 64,000-word visual vocabulary) |
| Learning signal | Denoising score-matching (EDM): recover the clean signal from a noised one | Next-token cross-entropy: predict the next code (exactly like an LLM) |
| Text conditioning | T5-XXL via cross-attention | T5 via cross-attention added to blocks |
| Training stages | Text2World pretrain → Video2World fine-tune (add observed frames) | Video-only next-token pretrain → add text for Video2World |
| Model sizes | 7B / 14B | 4B / 12B (5B / 13B for Video2World) |
| Strength | Highest visual fidelity | Speed + streaming; plugs into the LLM toolbox (KV-cache, real-time) |
| Weakness | Many denoising steps → slower | Heavy compression → artifacts → needs a diffusion decoder to repaint detail |

Table 2a. The two engines use different tokenizers, different losses, and different training stages — they are genuinely different models, not one model trained twice.

### So why keep both?

They are **complementary bets**. Diffusion is the **quality** champion; autoregressive is the **speed + LLM-integration** champion (it treats video like language — exactly the direction Cosmos 3 later embraced). The **diffusion decoder** exists only because the autoregressive path's aggressive discrete compression *"can sometimes lead to undesired distortions"* — so a small diffusion model cleans its output back up.

### Important: does this two-engine split apply to the OTHER families?

**No.** The diffusion-vs-autoregressive choice is a **Predict** thing. Transfer, Reason and the Tokenizer each use the one technique that fits their job:

| Family (gen 1) | Diffusion? | Autoregressive? | Core technique |
|---|---|---|---|
| Predict 1 | Yes (7B/14B) | Yes (4B/12B) | BOTH — the only two-engine family |
| Transfer 1 | Yes | No | diffusion DiT + ControlNet (wraps Predict's diffusion) |
| Reason 1 | No | Yes — over TEXT tokens | decoder-only VLM (+ SFT/RL); "distinct from diffusion" |
| Tokenizer 1 | No | No | autoencoder — outputs continuous (→diffusion) & discrete (→AR) tokens |

Table 2c. Only Predict has the two-engine split. Note "autoregressive" means different things: Predict predicts the next discrete VIDEO token (future frames); Reason is a normal LLM/VLM predicting the next TEXT token. In Cosmos 3 these become the two towers — generator (diffusion, Predict+Transfer) and reasoner (autoregressive-text, Reason).

## Why each new generation? (the problem it solved)

A fair question: was each new version just a **bigger model on more data**? **No.** Scale grew every time, but it was the *enabler*, not the headline — each step introduced a distinct architectural or algorithmic idea to fix a concrete problem.

| Step | Key problem with the previous version | How they solved it | Real novelty (beyond size/data) |
|---|---|---|---|
| **1 → 2** | Too slow, too many separate engines, visible hallucinations — a research platform, not a daily tool. | Commit to the diffusion path; add sparse attention; engineer better quality/control; add small fast variants; add action-conditioned post-training. | Sparse attention (≈2.6× faster) + action conditioning. |
| **2 → 2.5** | Three separate task-models to juggle; a T5 text encoder that doesn't "understand" physics; short, single-camera video. | Unify the 3 tasks into ONE flow-based model; replace T5 with the Cosmos-Reason1 VLM as encoder; add RL post-training + model merging; extend to 30s, multi-camera. | Task unification + flow matching + reasoning-VLM encoder + RL. |
| **2.5 → 3** | Still only a video predictor — can't natively reason, hear, or act; understanding and generation were separate models. | One mixture-of-transformers omnimodel: a reasoning transformer + an expert generation transformer; reasons before it generates; adds audio + action. | New architecture (MoT) + new modalities (audio/action) + merging understand/generate/act. |

Table 2b. Read the rightmost column: every jump is a genuine idea, not a scale-up. 1→2 buys speed; 2→2.5 buys understanding + simplicity; 2.5→3 buys a whole new capability.

### The one-sentence "why"

- **1→2:** "Make it fast and reliable enough to actually use."
- **2→2.5:** "Make it one model that truly understands the request."
- **2.5→3:** "Stop just watching the world — reason about it, hear it, and act in it."

## The data: scale and diversity

A WFM is only as good as the video it learns from. Cosmos 1 curated **~100 million clips** out of **20 million hours** of raw video, deliberately balanced across nine kinds of physical scene so the model isn't biased toward one (e.g. only driving). Later generations grew the data by orders of magnitude.

### Cosmos 1 training-data mix (by category)

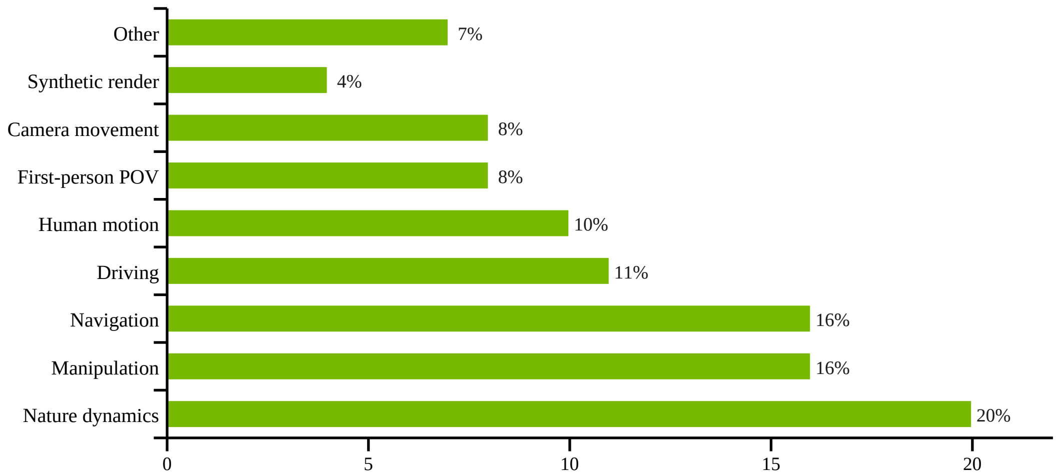



Figure 4. Diversity is intentional: nature, robot manipulation, and navigation dominate, but driving, human motion and synthetic data are all represented. This breadth is what lets one pretrained model be fine-tuned for many downstream robots/vehicles.

### Data scale across generations

| Generation | Pretraining data | Notes |
|---|---|---|
| Cosmos 1 | ~100M clips (from ~20M hours) | 9 balanced categories; ~30% removed by dedup |
| Cosmos 2 | Cosmos 1 data + action datasets | Bridge, AgiBotWorld, GR00T Dreams for post-training |
| Cosmos 2.5 | 200M curated high-quality clips | + domain post-training data (auto, robotics) |
| Cosmos 3 | ~20 trillion tokens | ~1B images, ~400M videos, + audio + action trajectories |

Table 2. The jump from “100M clips” to “20T tokens” reflects Cosmos 3 also ingesting text, audio and robot/human action data — not just video. **Caveat:** Cosmos 3's exact counts (20T tokens, ~1B images, ~400M videos) come from press coverage; NVIDIA's official release states only “billions of samples across text, image, video, sound and action.”

### The key insight: each generation adds a new TYPE of data

The data story is **qualitative**, not just “more”. Each generation introduces a new *kind* of data, and that new kind is exactly what unlocks the next capability:

- **Cosmos 1 — broad video** (9 balanced categories) → learns **general physics**.
- **Cosmos 2 — + action-labelled robot data** (Bridge, AgiBotWorld, GR00T Dreams) → learns **“this action causes that outcome.”**
- **Cosmos 2.5 — + multi-camera / domain data** (7-cam driving, 3-cam robotics) → learns **deployment realism and multi-view consistency.**
- **Cosmos 3 — + audio and action trajectories** (from humans and robots) → enables the **omnimodel** that can hear and act, not just see.

## Model sizes & key hyperparameters

“Parameters” (B = billion) are the model's adjustable knobs — more usually means smarter but slower/heavier. Cosmos offers small *and* large versions so you can trade quality for speed.

### Every model variant at a glance

| Generation | Variants (parameters) | Helper models |
| --- | --- | --- |
| Cosmos 1 | Diffusion 7B & 14B; Autoregressive 4B & 12B (5B/13B for Video2World) | Tokenizer 77–105M; Prompt-upsampler 12B; Diffusion-decoder 7B |
| Cosmos 2 | 0.6B (Text2Image), 2B, 14B | Sparse-attention inference path |
| Cosmos 2.5 | 2B & 14B (each: pre-trained + post-trained); auto 7-cam & robot 3-cam variants | Cosmos-Reason1 as text encoder |
| Cosmos 3 | Super (max accuracy), Nano (sub-second), Edge (local, soon) | Built-in reasoning transformer |

### Cosmos 1 diffusion hyperparameters (the most fully documented)

| Setting | 7B model | 14B model |
| --- | --- | --- |
| **Transformer layers** | 28 | 36 |
| **Model (hidden) dimension** | 4,096 | 5,120 |
| **Attention heads** | 32 | 40 |
| **AdaLN-LoRA rank** | 256 | 256 |
| **Learning rate** | 2^-15 | 2^-16 |
| **Optimizer** | AdamW (beta1=0.9, beta2=0.99), weight decay 0.1–0.2 | |
| **Warmup** | 2,500 iterations, linear | |
| **Text encoder** | T5-XXL, 512 tokens via cross-attention | |
| **Resolution schedule** | 512p (57 frames) → 720p (121 frames) | |
| **Context length @720p** | 56,320 tokens | |
| **Diffusion formulation** | EDM denoising score-matching; QK-RMSNorm for stability | |
| **Compute** | ~10,000 H100 GPUs, ~3 months | |

Table 3. NVIDIA published far more hyperparameters for Cosmos 1 than for the closed-er later releases; 2.5 and 3 reuse the same design DNA (3D-RoPE positions, AdaLN-LoRA, a causal tokenizer) at larger scale.

## Inputs and outputs: the “data contract” (DTO)

Think of a **DTO** (data-transfer object) as the exact shape of what you hand the model and what it hands back. This shape widened a lot across generations.

| Generation | INPUT (what you provide) | OUTPUT (what you get back) |
|---|---|---|
| Cosmos 1 (diffusion) | Text → T5-XXL embedding (512 tokens) + optional conditioning frames + fps + frame-count | Continuous latent video → decoded to RGB frames |
| Cosmos 1 (autoregressive) | Discrete video tokens (DV 8×16×16) + T5 text embedding | Next discrete tokens → diffusion-decoder → RGB frames |
| Cosmos 2 | Text, or image, or video + fps | RGB video at 480p / 704p, 10–16 fps |
| Cosmos 2.5 | Text / image / video, encoded by Cosmos-Reason1 | Up to 30s video; optional multi-camera (3–7 synced views) |
| Cosmos 3 | Text + image + video + ambient audio + action trajectories | Video + action trajectory + audio, preceded by explicit reasoning |

Table 4. The input distribution broadens from “text + frames” (Cosmos 1) to a full five-modality stream including audio and actions (Cosmos 3). The output likewise grows from “a short video” to “video + the physical actions a robot should take”.

### Why the input distribution matters

- **Cosmos 1–2:** trained mostly on curated *internet-style* video across 9 categories — great general physics, but you fine-tune for your specific robot.
- **Cosmos 2.5:** 200M *physical-AI-focused* clips + domain post-training (driving, manipulation) shift the distribution toward real deployment.
- **Cosmos 3:** adds *action trajectories from real humans and robots* and *audio* to the training mix — so the input/output distribution now includes the robot's own behaviour, not just what a camera sees.

## Key results: inference speed

Speed is the headline practical difference. Faster generation = more simulations per day = faster robot/AV development. Each generation attacked latency differently.

| Generation | Speed lever | Reported result |
|---|---|---|
| Cosmos 1 | Causal, factorized tokenizer | Up to 12× faster encode/decode vs. prior tokenizer (CogVideoX) |
| Cosmos 2 | Sparse attention in the DiT | Up to 2.6× end-to-end inference speedup |
| Cosmos 2.5 | Flow-based sampling (fewer steps) + smaller 2B option | Faster, more resource-efficient than 2.x diffusion |
| Cosmos 3 | Dedicated Nano variant | High-quality video + action reasoning in a fraction of a second |

Table 5. Two concrete, comparable speedups are published: Cosmos 1's tokenizer (up to 12×) and Cosmos 2's sparse attention (up to 2.6×). 2.5 and 3 emphasise efficiency via flow sampling and a sub-second Nano tier rather than a single headline multiplier.

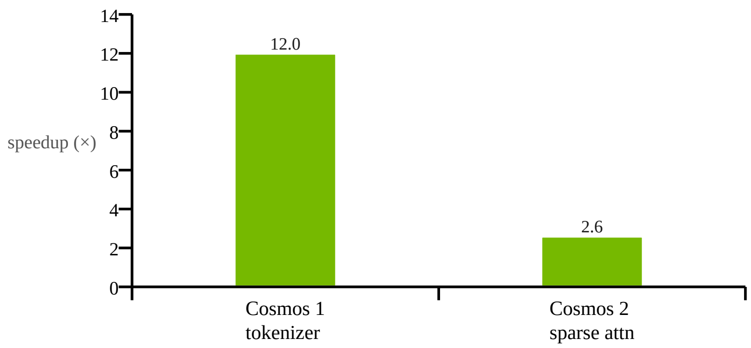



Figure 5. The two directly-quoted “× speedup” numbers. (Different things are being sped up — tokenizer vs. whole pipeline — so this compares *magnitude of improvement*, not absolute latency.)

### The quality/speed trade-off in one sentence

**Diffusion** (Cosmos 1–2) = many denoising steps = top quality but slower; **autoregressive** (Cosmos 1) = stream tokens = fast but rougher; **flow-based** (Cosmos 2.5) = fewer steps for similar quality; **Nano** (Cosmos 3) = engineered for sub-second responses.

## Zoom-in: what is “sparse attention”? (the Cosmos 2 trick)

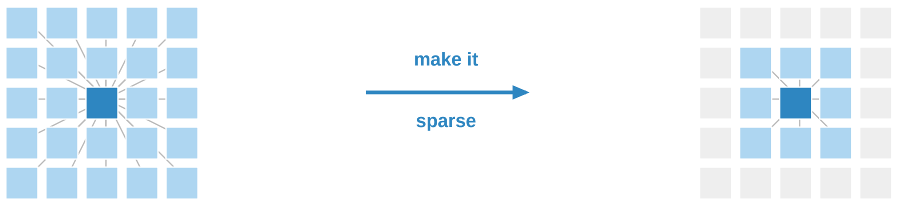
Cosmos 2's headline speedup (up to 2.6×) comes from **sparse attention**. Here is the idea with no maths.

### The problem: “attention” is very expensive for video

Inside a Transformer, every token (think: every little patch of the video) **looks at every other token** to decide what is relevant. That is called **full attention**. A short 720p video is **tens of thousands of patches** (frames × height × width), and full attention compares *every patch with every other patch* — so the cost grows **quadratically**: double the patches → **4×** the work. For video, this one step dominates the whole generation time.

### The insight: most of those comparisons are wasted

A patch in the top-left of frame 1 has almost nothing to do with a patch in the bottom-right three seconds later. What actually matters to a patch is its **neighbours** — nearby in space, and a frame or two before/after.




**FULL attention**
1 patch → ALL others (wasteful)

**NEIGHBOURHOOD (sparse) attention**
1 patch → only nearby patches (cheap)

Figure 5a. Left: every patch attends to all others (lines everywhere). Right: each patch attends only to its local window — the grey patches are skipped. Cosmos 2 (via the NATTEN library) drops up to 98% of the connections, keeping only the ~2% that matter.

### The fix: Neighbourhood Attention (NATTEN)

Instead of attending to **all** patches, each patch attends only to those in a small **local window** around it (in space and time). Cosmos 2 pushed this so **up to 98% of the attention connections are dropped** (sparsity raised “from 50% to 98%”) — only the most relevant ones are computed.

- **Result:** far fewer calculations → **1.7×–2.6× faster** at 720p, with quality essentially preserved (the dropped links were the unimportant ones).
- **Bonus:** it is an inference-time efficiency change, not a bigger model — related “Generalized Neighbourhood Attention” can even be plugged into existing models for 28–46% speedups **without any retraining**.

*Analogy: full attention = everyone in a stadium trying to talk to everyone else at once (chaos, won't scale). Neighbourhood attention = each person only talks to their own row and the rows just ahead/behind — you keep all the context that matters, and it scales.*

# Zoom-in: what is “flow matching”? (the Cosmos 2.5 trick)

Cosmos 2.5 switched from **diffusion** to a **flow-based** model. Here is what that means, again with no maths.

## Diffusion takes a wandering path

Recall the diffusion “sculptor”: it turns noise into video with **many tiny denoising steps**. The path it follows from pure noise to a finished video is **curved and wandering**, so it needs *lots* of small steps (often 30–50+) to stay on track. Each step is a full pass through a huge model — so many steps = slow.

## Flow matching learns a straight “current” instead

Picture noise on the left and real video on the right. Flow matching teaches the model a **direction to move** at every point — “which way, and how fast, to get from noise toward data”. It trains by connecting each noise sample to a real sample with a **straight line** and learning to follow it. Because the target paths are **nearly straight**, you can take **big steps** and still land in the right place — so you need **far fewer steps**.

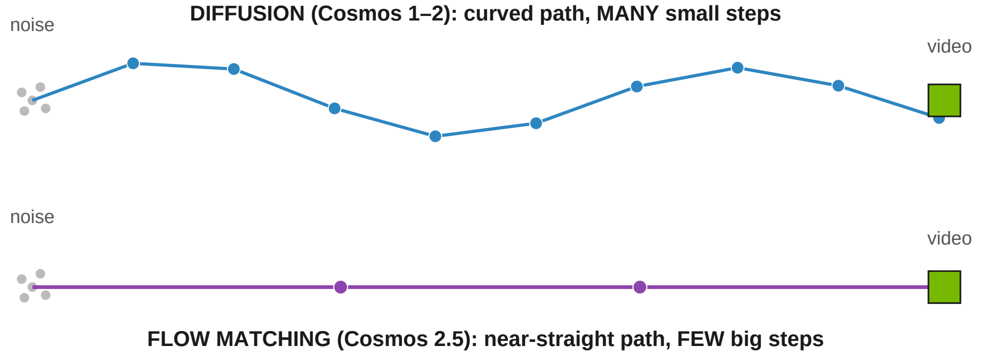



Figure 5b. Same start (noise) and destination (video). Diffusion zig-zags there in many small steps; flow matching learns a near-straight route it can cover in a few big steps.

## Why Cosmos 2.5 switched to it

- **Fewer sampling steps** → **faster & more resource-efficient** — important for 30-second, multi-camera video.
- **A cleaner, more stable training objective** (simply “match the direction”), which made it easier to build **one unified model** for Text/Image/Video-to-World instead of three.

*Analogy: diffusion is a windy mountain road with dozens of turns (many careful steps); flow matching is a straight highway to the same town — fewer, bigger moves to arrive.*

## Deep-dive: how Cosmos 3 fuses 5 modalities (incl. audio)

Cosmos 3 must ingest AND produce **five** very different data types — text, image, video, **audio**, **action** — and reason about them together (text is symbols, video is pixels, audio is a waveform, action is control vectors). How do you put all of that into **one** model? Three ingredients:

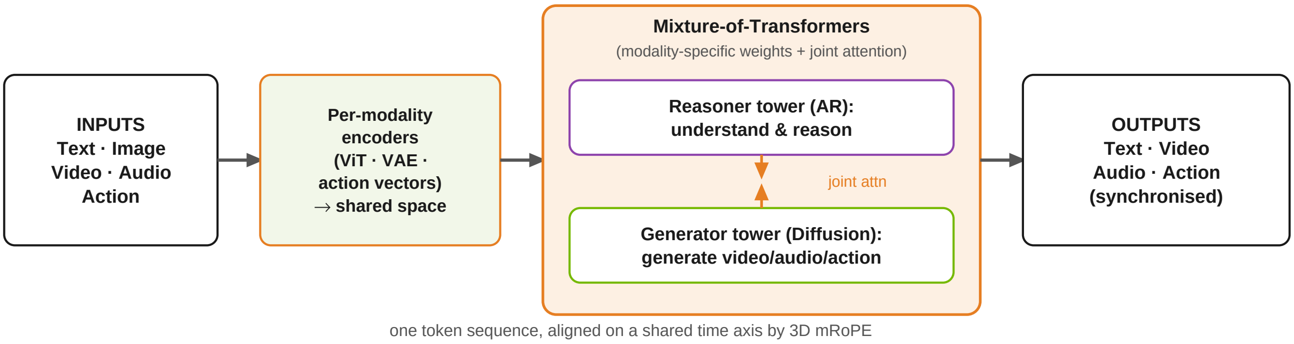



Figure 5c. Each modality is encoded into a shared space, concatenated into one sequence, and processed by a two-tower Mixture-of-Transformers: separate weights per modality/tower, but one global attention so audio, video, text and action all "see" each other.

### 1) A dedicated encoder per modality → one shared space

| Modality | How it enters the shared space |
|---|---|
| Text | Token embeddings (like an LLM) |
| Image / Video | A ViT for understanding; a VAE for generation |
| Audio | A VAE encodes the sound, then a linear projection maps audio tokens into the hidden dimension |
| Action | Domain-aware action vectors (robot / AV control) |

### 2) One sequence + global attention, aligned by 3D mRoPE

All tokens are concatenated into **one sequence** with **global self-attention** — an audio token can attend to a video token can attend to an action token. A unified **3D mRoPE** puts video, audio and action tokens **on one shared temporal axis** — literally how the model knows which sound goes with which frame and which action.

### 3) Mixture-of-Transformers — the "one group" mechanism

In a plain transformer all modalities fight over the same weights. **MoT gives each modality/tower its own weights** (feed-forward, attention projections, layer-norms) **while sharing global attention** — specialisation without fragmentation. The **Reasoner** (AR = understand) and **Generator** (diffusion = create) towers use separate parameters but **joint attention**; the reasoner can run alone, generation activates both. Sizes: **Nano 16B** (8B+8B), **Super 64B** (32B+32B).

### Managing the audio dataset specifically

- Audio is **ambient sound paired with video**, kept **temporally synchronised** and VAE-encoded; aligned to video/action via the shared mRoPE time axis.
- Curated through the same **multi-stage pipeline** (filter + quality review) as the rest of the data, from NVIDIA-owned + commercially-permissive sources.
- NVIDIA explicitly lists **"inaccurate sound–video alignment"** as a known failure mode — audio↔video timing is the central hard problem (and why mRoPE matters). *Granular audio-dataset counts are not publicly disclosed.*

# Pretraining and post-training: SFT and GRPO

Modern Cosmos models are built in two phases. **Pretraining** soaks up broad knowledge from massive data. **Post-training** then sharpens behaviour with curated examples (**SFT**) and reinforcement learning (**GRPO**). This recipe is clearest in **Cosmos-Reason1** — the reasoning brain that powers Cosmos 2.5's text understanding and the spirit of Cosmos 3.

## The 4-stage training pipeline (Cosmos-Reason1)

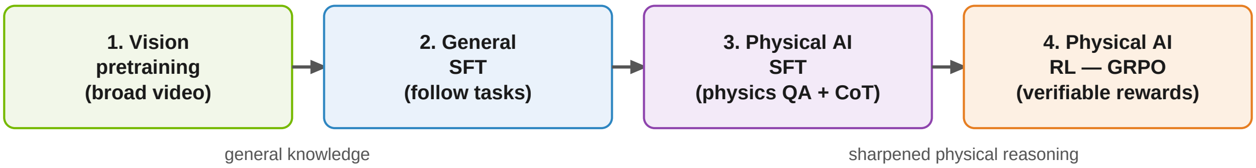



Figure 6. Knowledge first (stages 1–2), then physical-world specialisation (stage 3), then reinforcement learning to make the reasoning reliable (stage 4).

## What happens in SFT (stage 3)

- Curated **question** → **answer** examples for physical common sense (space, time, basic physics) and embodied decisions (what should the robot do?).
- Built from real robotics/AV datasets: **BridgeData V2, RoboVQA, AgiBot, HoloAssist**, and driving data.
- The team hand-picked **~200–250 examples per data source**, turned them into multiple-choice questions, and removed ambiguous ones — quality over quantity.
- Answers include **chain-of-thought** (the model explains its reasoning step by step).

## What happens in GRPO (stage 4) — in plain words

**GRPO** (Group Relative Policy Optimization) is reinforcement learning *without* a separate "judge" model. For each question the model produces **several attempts**; because these questions have a **verifiable correct answer**, each attempt can be auto-graded. Attempts that score **above the group average** are reinforced; below-average ones are discouraged. A **reference model** and **reward normalization** keep the updates stable so the model improves without "drifting". The payoff: noticeably better physical reasoning.

## Key experimental results

### Cosmos-Reason1: bigger model & RL both help

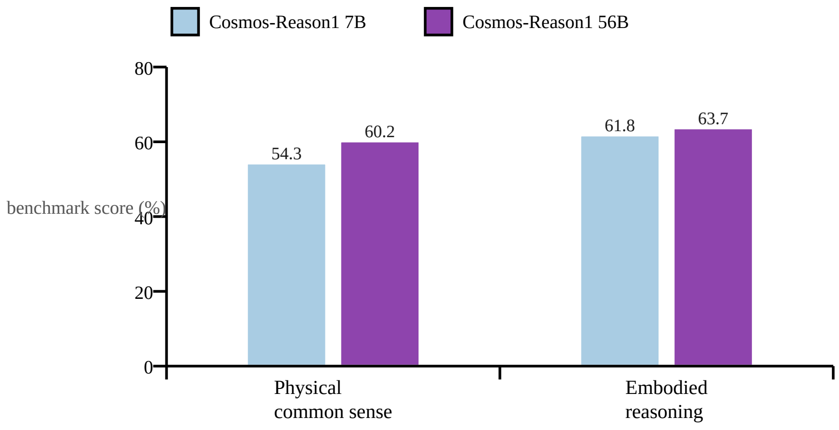



Figure 7. The 56B model beats the 7B on both physical-common-sense and embodied-reasoning benchmarks. Embodied reasoning improved by ~+10–11 points over the next-best prior model. **Source note:** numbers are from NVIDIA's Cosmos-Reason1 research page (released 7B checkpoint). An earlier arXiv v1 of the paper reported an *8B* model with somewhat different scores — see the verification page.

### Reinforcement learning (GRPO) lifts reasoning further

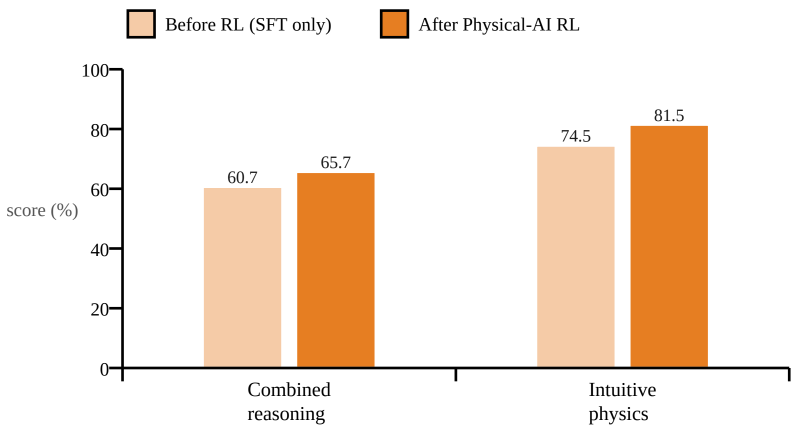



Figure 8. Adding the GRPO reinforcement-learning stage raised the 7B model by +5.0 points (combined reasoning) and +7.0 points (intuitive physics) — evidence that post-training, not just size, drives physical reasoning. **Source note:** research-page figures; arXiv v1 reports different magnitudes (e.g. intuitive physics 65.7→68.7). The *direction* — RL helps — is consistent across both.

### Generation-quality results (Cosmos 2.5 & 3)

- **Cosmos 2.5** post-trained **14B** scores **0.810** on PAI-Bench Image2World (matching the strong Wan2.2-A14B baseline).
- **Cosmos 2.5** autonomous-driving multiview model: up to **2.3×** better FVD/FID (video realism metrics) versus its predecessor.
- **Cosmos 3** ranks **#1 among open models** on Artificial Analysis, Physics-IQ, PAI-Bench and R-Bench (world generation); RoboLab and RoboArena (action policy); and VANTAGE-Bench and TAR (vision understanding).

Note: FVD/FID measure how realistic/consistent generated video is (lower is better, so a 2.3× improvement is large). PAI-Bench scores world-generation quality (higher is better).

## Verification & source audit

Every quantitative claim in this guide was cross-checked against primary sources (NVIDIA papers, research pages, and the official Cosmos 3 release). Of 20 checked claims, **18 verified** against a primary/official source; **2 carry caveats** (marked "!"). No factual errors were found in any claim that could be checked against a primary source.

| # | Claim in this guide | Checked against | Verdict |
|---|---|---|---|
| 1 | Cosmos 1 data mix: 20/16/16/11/10/8/8/4/7% | arXiv 2501.03575 | OK — exact |
| 2 | ~100M clips (10^8) from ~20M hours of video | arXiv 2501.03575 | OK — exact |
| 3 | Diffusion 7B = 28L / 4096 / 32 heads; 14B = 36L / 5120 / 40 | Table 11 | OK — exact |
| 4 | AdaLN-LoRA rank 256; LR 2^-15 / 2^-16; context 56,320 | arXiv 2501.03575 | OK — exact |
| 5 | Autoregressive 4B / 12B (+ 5B / 13B Video2World) | arXiv 2501.03575 | OK — exact |
| 6 | Trained on ~10,000 H100 GPUs for ~3 months | arXiv 2501.03575 | OK — exact |
| 7 | Tokenizer up to 12× faster; sparse attn (NATTEN) 50→98% sparsity, 1.7–2.6× | arXiv 2501.03575; arXiv 2504.16922 | OK |
| 8 | Cosmos 2: 0.6B/2B/14B; 480/704p; flow matching in 2.5 | GitHub + NVIDIA blog | OK |
| 9 | Cosmos 2.5: flow; Reason1 encoder; 2B/14B; 200M clips; 30s; PAI-Bench 0.810; 2.3× FVD/FID | NVIDIA research page | OK |
| 10 | GRPO post-training; 4-stage pipeline; verifiable rewards | arXiv 2503.15558 | OK |
| 11 | Cosmos-Reason1 = 7B & 56B with the Fig 7–8 scores | Research page vs arXiv v1 | CAVEAT — see below |
| 12 | Cosmos 3: mixture-of-transformers omnimodel; Super/Nano/Edge; #1 on open leaderboards | NVIDIA newsroom (Jun 2026) | OK |
| 13 | Cosmos 3 data = 20T tokens, ~1B images, ~400M videos | Press coverage (not NVIDIA) | CAVEAT — see below |
| 14 | Platform families: Predict / Transfer / Reason + Tokenizer / Curator / RL / Guardrails | Cosmos docs + DeepWiki | OK |
| 15 | Predict 1 variants (diffusion 7/14B, AR 4/12B, 5/13B V2W, upsampler, decoder) | arXiv 2501.03575 + GitHub | OK |
| 16 | Transfer = (Multi-)ControlNet on Predict; controls depth/edge/seg/LiDAR/HD-map | Cosmos docs + GitHub | OK |
| 17 | Transfer 2.5 built on Predict 2.5; 2B is 3.5× smaller than Transfer1-7B; control blocks distributed 1 per 7 (vs at start) | NVIDIA research page | OK — exact |
| 18 | Reason 1 = hybrid Mamba-MLP-Transformer; 16K context; SFT + RL | arXiv 2503.15558 (html) | OK — exact |
| 19 | Reason 2 = 2B/8B; 256K context (up from 16K); OCR + 2D/3D localization; #1 open | NVIDIA/HF Reason 2 blog | OK — exact |
| 20 | Cosmos 3 = two-tower MoT (Nano 16B = 8+8, Super 64B = 32+32); ViT/VAE encoders; 3D mRoPE | NVIDIA/HF Cosmos 3 blog | OK |

Table 6. Claims audit. Green = verified against a primary source; amber = caveat.

### The two caveats, in detail

- **! Cosmos-Reason1 size & scores (claim 11):** NVIDIA's research page describes a released **7B** model with the scores plotted in Figures 7–8. The **arXiv v1** of the paper instead describes an **8B** model and reports different numbers (e.g. physical common sense 52.3% vs 54.3%; intuitive-physics RL gain 65.7→68.7 vs 74.5→81.5). This is a paper-revision difference. This guide uses the **research-page (7B)** figures because that matches the publicly released checkpoint — but treat the exact decimals as version-dependent. The qualitative conclusions (bigger model helps; RL helps) hold in both.
- **! Cosmos 3 data scale (claim 13):** the "20 trillion tokens / ~1B images / ~400M videos" figures come from **press coverage** of the launch. NVIDIA's own newsroom release only says "billions of samples across text, image, video, sound and action." Treat the precise counts as **unconfirmed by NVIDIA**.

**Bottom line:** the guide is accurate where primary sources exist. The only soft spots are (a) decimal-level benchmark numbers for Cosmos-Reason1, which differ between paper versions, and (b) Cosmos 3's exact data counts, which NVIDIA has not officially published.

## Putting it all together

The four generations trace a clear arc:

- **Cosmos 1 (Toolbox):** two engines (diffusion + GPT), a strong causal tokenizer, T5 text, 9-category data, fully documented hyperparameters. Maximum flexibility.
- **Cosmos 2 (Polish):** diffusion-only, sparse attention for up to 2.6× speed, action-conditioned post-training. Same idea, executed better.
- **Cosmos 2.5 (Unify):** one flow-based model replaces three; Cosmos-Reason1 reads the prompt; 200M clips; 30s multi-camera video; RL + model-merge post-training.
- **Cosmos 3 (Leap):** mixture-of-transformers omnimodel; reasons before generating; adds audio and actions; ~20T-token training; Super/Nano/Edge tiers; #1 open model on many leaderboards.

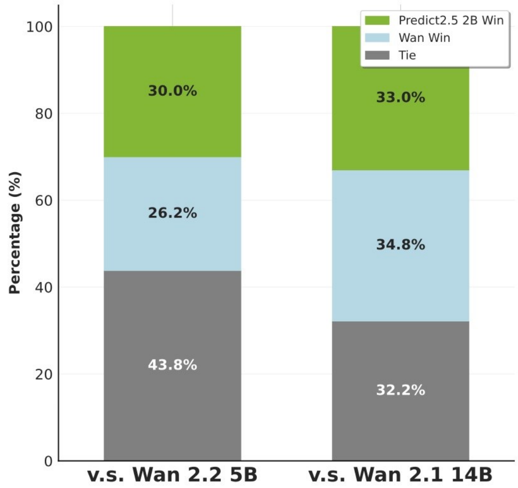
### The single sentence to remember

Cosmos went from **“many tools that dream video”** (1) → **“the best tool, polished”** (2) → **“one smart unified tool”** (2.5) → **“a reasoning, seeing, hearing, acting omnimodel”** (3).

# Appendix A — Example results (from NVIDIA's pages)

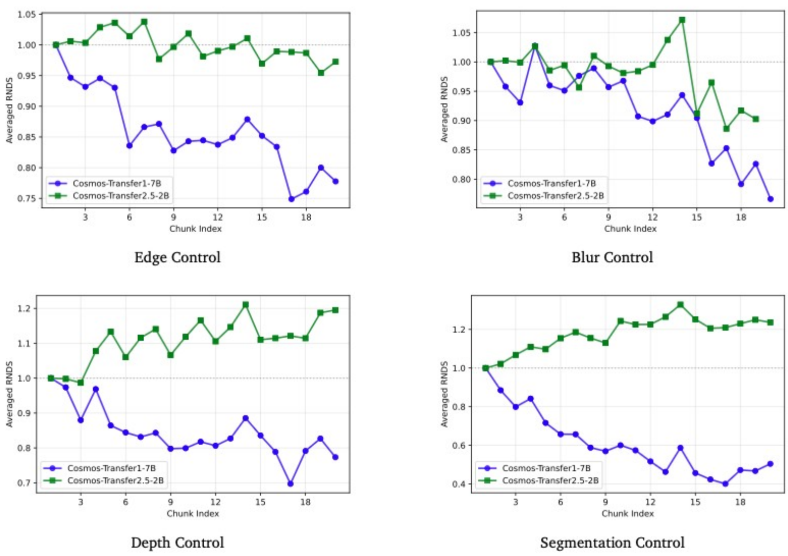
These are real result figures and example inputs taken from NVIDIA's official Cosmos pages, reproduced here for educational reference. © NVIDIA; see the link under each image.

## A1 · Cosmos Predict 2.5 — human-preference win rate vs. Wan baselines



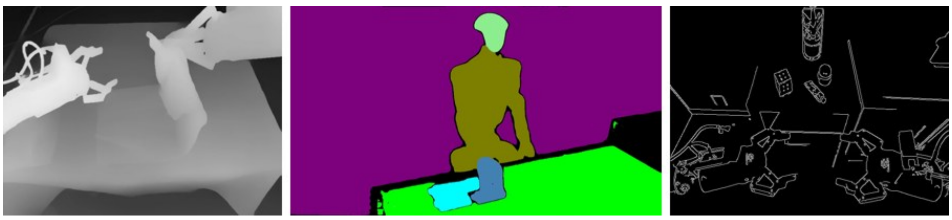

Predict 2.5 (2B) “win / tie / lose” against Wan 2.2 (5B) and Wan 2.1 (14B) in human preference — a small model competitive with much larger baselines. Source: research.nvidia.com/labs/cosmos-lab/cosmos-predict2.5/

## A2 · Cosmos Transfer 2.5 (2B) vs. Transfer 1 (7B) — quality across long videos




Averaged RNDS (higher = better) across video “chunk index” for Edge / Blur / Depth / Segmentation control. Transfer 2.5-2B (green) stays high while Transfer 1-7B (blue) degrades over long sequences — the 3.5× smaller model is also better. Source: research.nvidia.com/labs/cosmos-lab/cosmos-transfer2.5/ (Figure 8)

## A3 · What Cosmos Transfer ingests — example control inputs



Left→right: depth, segmentation and edge control maps. Cosmos Transfer turns structured inputs like these into photorealistic video (sim-to-real). Source: research.nvidia.com/labs/cosmos-lab/cosmos-transfer2.5/

# Appendix B — References & further reading

## Primary papers


• **Cosmos World Foundation Model Platform for Physical AI (Cosmos 1)**
https://arxiv.org/abs/2501.03575

• **Cosmos-Reason1: From Physical Common Sense to Embodied Reasoning**
https://arxiv.org/abs/2503.15558

• **Mixture-of-Transformers: A Sparse, Scalable Multi-Modal Architecture**
https://arxiv.org/abs/2411.04996

• **Generalized Neighborhood Attention (sparse attention / NATTEN)**
https://arxiv.org/abs/2504.16922


## NVIDIA research & model pages


• **Cosmos-Predict2.5**
https://research.nvidia.com/labs/cosmos-lab/cosmos-predict2.5/

• **Cosmos-Transfer2.5**
https://research.nvidia.com/labs/cosmos-lab/cosmos-transfer2.5/

• **Cosmos 3 technical report**
https://research.nvidia.com/labs/cosmos-lab/cosmos3/technical-report.pdf

• **Cosmos model families (docs / DeepWiki)**
https://deepwiki.com/nvidia-cosmos/cosmos-cookbook/3-cosmos-model-families


## Blogs & announcements


• **HF: Cosmos Predict 2.5 & Transfer 2.5**
https://huggingface.co/blog/nvidia/cosmos-predict-and-transfer2-5

• **HF: Cosmos Reason 2 brings advanced reasoning**
https://huggingface.co/blog/nvidia/nvidia-cosmos-reason-2-brings-advanced-reasoning

• **HF: Welcome Cosmos 3 for Physical AI**
https://huggingface.co/blog/nvidia/cosmos-3-for-physical-ai

• **NVIDIA blog: Develop Physical AI with Cosmos 3**
https://developer.nvidia.com/blog/develop-physical-ai-reasoning-world-and-action-models-with-nvidia-cosmos-3/

• **NVIDIA newsroom: Cosmos 3 launch (Jun 2026)**
https://nvidianews.nvidia.com/news/nvidia-launches-cosmos-3-the-open-frontier-foundation-model-for-physical-ai


## GitHub


• **cosmos-predict1 / predict2 / predict2.5**
https://github.com/nvidia-cosmos

• **cosmos-transfer1 / transfer2.5**
https://github.com/nvidia-cosmos

• **cosmos-reason1 / reason2**
https://github.com/nvidia-cosmos


 All numeric claims are quoted from the sources above; where sources disagree or NVIDIA has not published a figure, this is flagged on the verification page. Educational summary.